

《数据库系统原理》实验报告（实验四）					
题目：实验四：SQL 综合实验					
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实验环境： Docker OceanBase					
实验步骤及结果截图： 0. 创建数据库 cinema 并使用 <pre>obclient(root@sys)[(none)]> create database cinema; Query OK, 1 row affected (0.043 sec) obclient(root@sys)[(none)]> use cinema; Database changed -</pre> 1. 建表 Movie, Viewer, Watch <pre>obclient(root@sys)[cinema]> create table Movie (-> Movie_no varchar(10) primary key, -> Movie_name varchar(50), -> Director varchar(30), -> Rating float, -> End_date datetime ->); Query OK, 0 rows affected (0.065 sec) obclient(root@sys)[cinema]> create table Viewer (-> Viewer_no varchar(10) primary key, -> Viewer_name varchar(30), -> Age int ->); Query OK, 0 rows affected (0.040 sec) obclient(root@sys)[cinema]> create table Watch (-> Viewer_no varchar(10), -> Movie_no varchar(10), -> Watch_date datetime, -> primary key(Viewer_no, Movie_no), -> foreign key(Viewer_no) references Viewer(Viewer_no) on delete cascade, -> foreign key(Movie_no) references Movie(Movie_no) on delete cascade ->); Query OK, 0 rows affected (0.086 sec)</pre> 2. 插入样例数据 <pre>obclient(root@sys)[cinema]> insert into Movie values -> ('M001', '星际穿越', '克里斯托弗·诺兰', 9.3, '2024-05-01'), -> ('M002', '泰坦尼克号', '詹姆斯·卡梅隆', 9.1, '2024-04-10'), -> ('M003', '盗梦空间', '克里斯托弗·诺兰', 8.8, '2024-04-20'), -> ('M004', '科幻冒险之旅', '张三', 7.5, '2024-04-18'), -> ('M005', '爱情故事', '李四', 7.0, '2024-04-25'); Query OK, 5 rows affected (0.008 sec) Records: 5 Duplicates: 0 Warnings: 0 obclient(root@sys)[cinema]> insert into Viewer values -> ('V001', '李明', 25), -> ('V002', '王红', 30), -> ('V003', '张磊', 22), -> ('V004', '赵颖', 28), -> ('V005', '孙阳', 35); Query OK, 5 rows affected (0.010 sec) Records: 5 Duplicates: 0 Warnings: 0</pre>					

```
obclient(root@sys)[cinema]> insert into Watch values
-> ('V001', 'M001', '2024-03-15'),
-> ('V001', 'M002', '2024-03-20'),
-> ('V002', 'M002', '2024-03-25'),
-> ('V002', 'M003', '2024-04-01'),
-> ('V003', 'M001', '2024-04-05'),
-> ('V004', 'M002', '2024-04-12'),
-> ('V005', 'M003', '2024-04-14');
Query OK, 7 rows affected (0.010 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

3. 查询电影名称中包含“科幻”的电影信息，输出所有信息（包括电影名称、电影编号、导演、评分、电影停映日期），并按照评分降序排列。

```
obclient(root@sys)[cinema]> select * from Movie
-> where Movie_name like '%科幻%'
-> order by rating desc;
+-----+-----+-----+-----+-----+
| Movie_no | Movie_name      | Director | Rating | End_date          |
+-----+-----+-----+-----+-----+
| M004     | 科幻冒险之旅   | 张三    | 7.5    | 2024-04-18 00:00:00 |
+-----+-----+-----+-----+
1 row in set (0.002 sec)
```

4. 查询观看了电影名为“泰坦尼克号”的观众信息，输出该观众的编号、姓名和年龄，并按照观众编号升序排列。

```
obclient(root@sys)[cinema]> select distinct v.viewer_no, v.viewer_name, v.age
-> from viewer v
-> join watch w on v.viewer_no = w.viewer_no
-> join movie m on w.movie_no = m.movie_no
-> where m.movie_name = '泰坦尼克号'
-> order by v.viewer_no asc;
+-----+-----+-----+
| viewer_no | viewer_name | age |
+-----+-----+-----+
| V001     | 李明       | 25  |
| V002     | 王红       | 30  |
| V004     | 赵颖       | 28  |
+-----+-----+-----+
3 rows in set (0.021 sec)
```

5. 统计每个观众的观影信息，输出每个观众的编号、观看的电影名称和观看日期。

```
obclient(root@sys)[cinema]> select w.viewer_no, m.movie_name, w.watch_date
-> from watch as w
-> join movie as m on w.movie_no = m.movie_no
-> order by w.viewer_no;
+-----+-----+-----+
| viewer_no | movie_name      | watch_date          |
+-----+-----+-----+
| V001     | 星际穿越       | 2024-03-15 00:00:00 |
| V001     | 泰坦尼克号     | 2024-03-20 00:00:00 |
| V002     | 泰坦尼克号     | 2024-03-25 00:00:00 |
| V002     | 盗梦空间       | 2024-04-01 00:00:00 |
| V003     | 星际穿越       | 2024-04-05 00:00:00 |
| V004     | 泰坦尼克号     | 2024-04-12 00:00:00 |
| V005     | 盗梦空间       | 2024-04-14 00:00:00 |
+-----+-----+-----+
7 rows in set (0.014 sec)
```

6. 查询所有已停映电影的信息，输出观众编号、姓名、电影名称和观看日期，并按观看日期降序排列。

P.S.已停映电影指的是“现实日期”大于电影停映日期字段的电影，“现实日期”以4月15日为例。

```
obclient(root@sys)[cinema]> select v.viewer_no, v.viewer_name, m.movie_name, w.watch_date
-> from viewer v
-> join watch w on v.viewer_no = w.viewer_no
-> join movie m on w.movie_no = m.movie_no
-> where datediff(end_date, '2024-04-15') < 0
-> order by w.watch_date desc;
+-----+-----+-----+-----+
| viewer_no | viewer_name | movie_name | watch_date |
+-----+-----+-----+-----+
| V004      | 赵颖        | 泰坦尼克号 | 2024-04-12 00:00:00 |
| V002      | 王红        | 泰坦尼克号 | 2024-03-25 00:00:00 |
| V001      | 李明        | 泰坦尼克号 | 2024-03-20 00:00:00 |
+-----+-----+-----+-----+
3 rows in set (0.010 sec)
```

7. 查询观看了“星际穿越”但没有观看“盗梦空间”的观众信息，输出这些观众的编号，并按照编号升序排列。

```
obclient(root@sys)[cinema]> select distinct w.viewer_no
-> from watch as w
-> join movie as m on w.movie_no = m.movie_no
-> where m.movie_name = '星际穿越'
-> and
-> w.viewer_no not in
-> (select w2.viewer_no
-> from watch as w2
-> join movie as m2 on w2.movie_no = m2.movie_no
-> where m2.movie_name = '盗梦空间'
-> )
-> order by w.viewer_no;
+-----+
| viewer_no |
+-----+
| V001      |
| V003      |
+-----+
2 rows in set (0.019 sec)
```

8. 创建一个过程，使之能够实现如下功能：

9. 修改观影表，增加字段“重复观看状态”（字段名为“Repeat_state”），字段含义为表示观众是否多次观看该电影；

10. 并根据表中已有数据为该字段赋值（所赋的值与表定义时的数据类型保持一致即可，比如可以定义多次观看某电影的“重复观看状态”为 True，只看过一次某电影的“重复观看状态”为 False），要求使用 if 语句进行条件判断。

```
obclient(root@sys)[cinema]> delimiter $$
obclient(root@sys)[cinema]>
obclient(root@sys)[cinema]> create procedure update_repeat_state()
-> begin
->     declare v_no varchar(10);
->     declare m_no varchar(10);
->     declare record_count int;
->     declare done int default false;
->     declare cur cursor for
->         select distinct Viewer_no, Movie_no from Watch;
->     declare continue handler for not found set done = true;
->
->     alter table watch
->         add column Repeat_state varchar(10);
->
->     open cur;
->
```

```

-> read_loop: loop
->   fetch cur into v_no, m_no;
->   if done then
->     leave read_loop;
->   end if;
->
->   select count(*) into record_count
->   from Watch
->   where Viewer_no = v_no and Movie_no = m_no;
->
->   if record_count > 1 then
->     update Watch
->     set Repeat_state = 'true'
->     where Viewer_no = v_no and Movie_no = m_no;
->   else
->     update Watch
->     set Repeat_state = 'false'
->     where Viewer_no = v_no and Movie_no = m_no;
->   end if;
->
-> end loop;
->
-> close cur;
->
-> end $$
Query OK, 0 rows affected (0.057 sec)

obclient(root@sys)[cinema]>
obclient(root@sys)[cinema]> delimiter ;
obclient(root@sys)[cinema]>

```

执行过程

```

obclient(root@sys)[cinema]> call update_repeat_state;
Query OK, 1 row affected (0.267 sec)

```

```

obclient(root@sys)[cinema]> select * from watch;
+-----+-----+-----+-----+
| Viewer_no | Movie_no | Watch_date       | Repeat_state |
+-----+-----+-----+-----+
| V001      | M001     | 2024-03-15 00:00:00 | false        |
| V001      | M002     | 2024-03-20 00:00:00 | false        |
| V002      | M002     | 2024-03-25 00:00:00 | false        |
| V002      | M003     | 2024-04-01 00:00:00 | false        |
| V003      | M001     | 2024-04-05 00:00:00 | false        |
| V004      | M002     | 2024-04-12 00:00:00 | false        |
| V005      | M003     | 2024-04-14 00:00:00 | false        |
+-----+-----+-----+-----+
7 rows in set (0.001 sec)

```

11. 在 8-10 题的基础上，查询没有重复观看过电影的观众信息，输出观众姓名和编号。

```

obclient(root@sys)[cinema]> select distinct v.viewer_no,v.viewer_name
-> from viewer as v
-> join watch as w on v.viewer_no = w.viewer_no
-> where w.repeat_state = 'false';
+-----+-----+
| viewer_no | viewer_name |
+-----+-----+
| V001      | 李明        |
| V002      | 王红        |
| V003      | 张磊        |
| V004      | 赵颖        |
| V005      | 孙阳        |
+-----+-----+
5 rows in set (0.015 sec)

```

12. 修改电影表，在 Movie_name 列上增加唯一性索引 Movie_name_index，并按 Movie_name 降序排列。

```
obclient(root@sys)[cinema]> create unique index movie_name_index on movie(movie_name);
Query OK, 0 rows affected (0.475 sec)

obclient(root@sys)[cinema]> select * from movie order by movie_name desc;
+-----+-----+-----+-----+-----+
| Movie_no | Movie_name      | Director      | Rating | End_date      |
+-----+-----+-----+-----+-----+
| M001     | 星际穿越        | 克里斯托弗·诺兰 | 9.3    | 2024-05-01 00:00:00 |
| M002     | 泰坦尼克号      | 詹姆斯·卡梅隆   | 9.1    | 2024-04-10 00:00:00 |
| M003     | 盗梦空间        | 克里斯托弗·诺兰 | 8.8    | 2024-04-20 00:00:00 |
| M005     | 爱情故事        | 李四           | 7      | 2024-04-25 00:00:00 |
| M004     | 科幻冒险之旅    | 张三           | 7.5    | 2024-04-18 00:00:00 |
+-----+-----+-----+-----+-----+
5 rows in set (0.002 sec)
```

出现的问题:

1.在创建 procedure 过程时, begin 后面直接加上语句

alter table watch

add column Repeat_state varchar(10);

结果一直报语法错误

2.查找时出现多条重复的记录

解决方案:

1.查资料得知, procedure 里面的声明 declare 一定要在 begin 后的最开始加上

修改后没有语法错误

2.使用 distinct 去除重复的行